

Thm. Boundedness

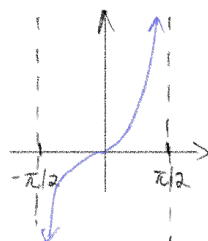
If f is continuous on $[a, b]$, then f is bounded above.

$$\hookrightarrow \exists N \in \mathbb{R} \text{ s.t. } f(x) \leq N \\ \forall x \in [a, b].$$

If f weren't continuous, then f might not be bounded.

If the interval were open

ex. $f(x) = \tan x$, not bounded



If the interval were not bounded (eg. $[a, \infty)$).

Cor. If f is cts. on $[a, b]$, then f is bounded below

$$\exists B \text{ s.t. } B \leq f(x) \quad \forall x \in [a, b].$$

Pf. Let $g(x) = -f(x)$

$$\Rightarrow g = (-1)f \quad \text{cts. on } [a, b].$$

Apply boundedness theorem on g .

$$\exists N \text{ s.t. } g(x) \leq N \quad \forall x \in [a, b].$$

$$\Leftrightarrow -f(x) \leq N \quad \forall x \in [a, b]$$

$$\Leftrightarrow -N \leq f(x) \quad \forall x \in [a, b]$$



Extreme Value Theorem

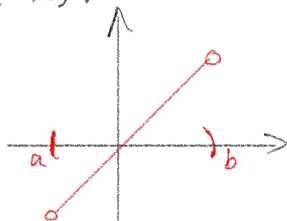
If f is cts. on $[a, b]$, then $\exists y \in [a, b]$

s.t. $f(x) \leq f(y) \quad \forall x \in [a, b]$

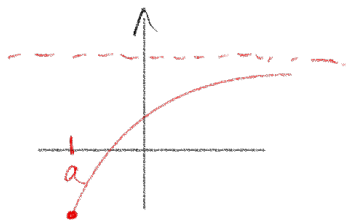
(ie. f is bounded above and its upper bound is achieved by f).

- If f were not cts. but bounded.

- If f were cts. and bounded but on (a, b) :



- If f cts. and bounded on $[a, \infty)$.



Cor. If f is cts. on $[a, b]$, then $\exists y \in [a, b]$ s.t. $f(y) \leq f(x) \quad \forall x \in [a, b]$.

Ex. ① Show that any $\alpha > 0$ has a square root

Solution. We'll use IVT on $f(x) = x^2 - \alpha$

Look at the interval $[0, \max\{\alpha, 2\}]$.

Then $f(0) = -\alpha < 0$

$$f(\max\{\alpha, 2\}) = \begin{cases} \alpha^2 - \alpha, & \text{if } \alpha \geq 2 \\ 4 - \alpha, & \text{if } \alpha < 2 \end{cases}$$

• $4 - \alpha > 0$ if $\alpha < 2$

• If $\alpha \geq 2$, need $\alpha^2 - \alpha > 0$

$$\alpha^2 - \alpha = \alpha(\alpha - 1).$$

Since $\alpha \geq 2$, $\alpha - 1 \geq 1$

So the product of positive # is positive.

So $[0, \max\{\alpha, 2\}]$ satisfies $f(a) < 0 < f(b)$.

By IVT, $\exists c \in [a, b]$ s.t. $f(c) = 0 \Rightarrow c = \sqrt{\alpha}$



② Prove that there exists some $x \in \mathbb{R}$ s.t.

$$x^{179} + \frac{163}{1+x^2 \sin x} = 119$$

Pf. Let $f(x) = x^{179} + \frac{163}{1+x^2 \sin x} - 119$

Then f is cts. ($1+x^2 \sin x \geq 1$).

$$\bullet f(0) = 0 + \frac{163}{1+0+0} - 119 > 0$$

$$\bullet f(-2) = (-2)^{179} + \frac{163}{\text{Something} \geq 1} - 119 < 0$$

$\text{but } \leq 163$

$\Rightarrow f(-2) < 0 < f(0)$, and f cts.

$\Rightarrow \text{IVT} \Rightarrow \exists c \in [-2, 0]$

s.t. $f(c) = 0$

③ If f cts. on $[a, b]$ and $f(x) \in \mathbb{Q}$, $\forall x \in [a, b]$

What can we say about f ?

Pf. We claim that f must be constant

Suppose for a contradiction that f is not constant.

So $\exists r, s \in \mathbb{Q}$ s.t. f takes on r, s .

Let $f(x) = r$, $x, y \in [a, b]$

$f(y) = s$

WOLOG, assume $x < y$, $r < s$

Let R be an irrational # besides s and r .

Let $g(x) = f(x) - R$

PF. We claim that f must be constant

Suppose for contradiction that f is not constant

So $\exists r, s \in \mathbb{Q}$ s.t. f takes on r, s .

- g is cts. on $[x, y]$
- $g(x) = f(x) - R = r - R < 0$

$$g(y) = f(y) - R = s - R > 0$$

By IVT, $\exists c \in [x, y]$ s.t. $g(c) = 0$.